

Daniel Lu

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Education

Carleton University, B.Sc. in Computer Science (Honours, Co-op)

Ottawa, Canada

AI & Machine Learning Stream - GPA: 3.9/4.0

Sep 2023 – Apr 2028

- **Coursework:** Supervised/Unsupervised Methods, Transformer Architectures, Language Modeling & Embeddings

Experience

Software Developer

May 2026 – Aug 2026

Shopify

Toronto, Canada

- Engineered core improvements to Shopify's Payments API to move checkouts-started-per-minute from around **20,000** toward the **42,000** target.
- Refactored the payment execution flow to optimize data access, clarify system boundaries, and improve resilience to transient failures, **reducing avoidable slowdowns during high-traffic periods**.

Software Developer

May 2025 – Aug 2025

TrendAI

Ottawa, Canada

- Pioneered TrendAI's **first** MCP-based digital twin prototype, architecting a Neo4j graph RAG pipeline for predictive infrastructure security analytics, directly securing C-suite approval for full-scale development.
- Improved system latency by **30%** and reduced monthly cloud costs by **18%** across **120+ active enterprise users** via AWS cloud infrastructure optimizations and Kubernetes Horizontal Pod Autoscaling.
- Engineered CI/CD pipelines for internal SDKs and database deployment systems serving **200+ developers** and **10,000+ stakeholders**, reducing release cycle duration by **65%** (**10 hours** → **under 4 hours**).

Software Developer

Jan 2025 – Apr 2025

Blackberry QNX

Ottawa, Canada

- Engineered core automation libraries in Python for QNX RTOS, expanding code coverage by from **60%** to **75%**.
- Reduced system validation cycle time by **19%** by engineering automated testing scripts in Python, C, and Bash for QNX RTOS build pipelines, cutting manual verification effort from **8 hours** to **6.5 hours** per release.

Projects

Beat Battle | [Website](#) | [GitHub](#)

- Engineered a competitive multiplayer music production platform on a stateless Bun + Elysia backend, using Redis pub/sub to synchronize **2000+ concurrent players** across **400+ live lobbies** at **sub-100ms** latency.
- Reached **150k+ accounts** and **1.2M+ unique site visits** within 13 days of launch with zero marketing spend, driven entirely by organic word-of-mouth and a rapidly growing **50,000-member** Discord community.

Mime | [GitHub](#)

- Bridged the visual-audio immersion gap in cross-lingual video calls by engineering a lightweight real-time 3D facial overlay that renders lip-synced translated speech with **91% accuracy**.
- Architected and trained a CNN-Transformer audio encoder on the 76-hour BEAT motion-capture dataset, regressing 52 ARKit facial blendshape coefficients at **0.0098 MSE**.
- Shipped a zero-client-modification pipeline compatible with all major video conferencing platforms, achieving **<250ms** end-to-end streaming latency and **100ms TTFT**.

Tobio | [Website](#) | [GitHub](#)

- Engineered a sensor-free 3D player tracking system from standard game footage using a monocular 3D reconstruction pipeline, deploying the product to **3 university teams + Brazilian men's national team** with no specialized hardware.
- Cut post-game analysis time by **85%** by piping a sliding-window action classifier into a vector-indexed RAG pipeline, enabling natural language querying of plays, rotations, and set patterns.
- Achieved **sub-centimeter tracking precision** on player and ball trajectories by pairing YOLOv11 detection with a ReID module to preserve player identity through occlusion, enabling continuous real-time 3D mapping across full match footage.

Technical Skills

Languages: Python, C/C++, Java, JavaScript, TypeScript, Rust, Ruby, SQL

Libraries and Frameworks: Hugging Face, PyTorch, NumPy, Scikit-learn, React, FastAPI

Platforms & Technologies: AWS, Docker, Kubernetes, Jenkins, Terraform, REST APIs, Git